

The role of the TMS parameters for activation of the corticospinal pathway to the diaphragm



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HIGHLIGHTS

- Biphasic- compared to monophasic-TMS requires significantly less charge and time to induce diaphragmatic motor-evoked potentials.
- Neuronavigated cortical mapping revealed similar best stimulation site of the diaphragm for monophasic- and biphasic-TMS.
- The TMS-stimulus modality significantly influences the reliability of the diaphragmatic MEPs.

ABSTRACT

Objective: The influence of the TMS-parameters on the efficacy and reliability to induce diaphragmatic motor-evoked potentials (diMEPs) has not been studied so far. Therefore, the objective of the present research is to probe the role of TMS- waveform (monophasic- [Mo] vs. biphasic-pulses [Bi]) and current direction (posterior-anterior [Pa] vs. anterior-posterior [Ap]) in the activation of the diaphragm.

Methods: Four different pulse-configurations (Mo-Ap, Mo-Pa, Bi-Ap, Bi-Pa) were applied by means of neuronavigated-TMS and surface MEP-recordings at relaxed end-expiration in 19 healthy subjects. The parameters resting motor threshold (RMT), diMEP-amplitude and -latency, as well as best stimulation site (motor hotspot) and central motor conduction time were studied. Diaphragm movements were simultaneously recorded via ultrasound. To control for possible signal contamination the MEPs of muscles neighboring the diaphragm were also obtained.

Results: The motor hotspots of the diaphragm showed similar spatial distribution for the Mo-Ap, Mo-Pa, Bi-Ap and Bi-Pa. The biphasic-pulses yielded significantly lower RMTs and higher diMEP-amplitudes as the monophasic-pulses. Anterior to posterior oriented Bi- and Mo-pulses evoked significantly shorter diMEP-latencies than the posterior-anterior oriented ones.

Conclusions: The present research demonstrates that biphasic- as compared to monophasic-pulses require significantly less charge and time for inducing diMEPs.

Significance: The biphasic-TMS is best suited for the demanding stimulation of the diaphragm.

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1. Introduction

Transcranial magnetic stimulation (TMS) of the cortical representation of the breathing muscles and corresponding registration of motor-evoked potentials (MEPs) allows noninvasive inspection of the integrity of the cortico-diaphragmatic pathway (Maskill

et al., 1991) for both clinical (e.g. multiple sclerosis (Lagueny et al., 1998), amyotrophic lateral sclerosis (Miscio et al., 2006; Similowski et al., 2000), tetraplegia (Lissens and Vanderstraeten, 1996), stroke (Khedr et al., 2000; Similowski et al., 1996a), chronic obstructive pulmonary disease (Hopkinson et al., 2012), featuring patients for phrenic nerve pacing (Similowski et al., 1996c)) and

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research applications (Dayer et al., 2007; Demoule et al., 2003; Sharshar et al., 2003, 2004b; Verin et al., 2004).

Several features of the utilized magnetic stimulator as well as the physical features of the delivered pulses per se have an impact on the mechanisms of MEP generation. Thus, the efficiency of different magnetic stimulation devices and the effects of different waveforms on the MEPs were increasingly explored in the last two decades with respect to single-pulse TMS (Kammer et al., 2001; Sommer et al., 2006; Stephani et al., 2016) and repetitive TMS (Antal et al., 2002; Sommer et al., 2002; Tings et al., 2005). It has been repeatedly shown that monophasic pulses require higher stimulation-intensity than biphasic pulses to elicit MEPs of the hand-muscles (Sommer et al., 2006, 2002; Stephani et al., 2016). However, single-pulse TMS of the diaphragm, with some exceptions (Miscio et al., 2006), has been mainly applied by devices providing monophasic pulses. In fact, often maximal stimulation output or intensities closer to MSO were required to elicit diaphragmatic MEPs (Davey et al., 1996; Kawaguchi et al., 2000; Lefaucheur and Lofaso, 2002; Lissens and Vanderstraeten, 1996; Maskill et al., 1991; Murphy et al., 1990; Rosler et al., 1994; Welch et al., 2020; Zifko et al., 1996). On the other hand, particularly for reasons of safety (especially when applying rTMS) accurately dosing the stimulation-intensity is of greatest importance in both research and clinical TMS studies (Rossi et al., 2021).

In times with increasing number of patients with respiratory failure, tentative diagnostic (in terms of single-pulse TMS) and innovative therapeutic approaches (by means of optimization of rTMS-stimulation parameters) warrant more detailed examination. Therefore, the objective of the present study is to quantify the effects of TMS pulse-configuration, in terms of waveform (monophasic vs. biphasic pulses) and current direction (posterior-anterior vs. anterior-posterior) on the resting motor threshold and the amplitude and latency of the diaphragmatic motor-evoked potentials (diMEPs) at relaxed end-expiration.

2. Materials and methods

2.1. Subjects

Twenty right-handed subjects aged 18 or older (mean age 27 (19–38) years) participated in this study. Exclusion criteria were previous or ongoing diseases of the CNS, implants or metallic material inside the head or neck, cardiac pacemaker, pregnancy, or lactation. The study was approved by the Ethics Committee of the University of Göttingen and was executed according to the declaration of Helsinki, including the amendment from Edinburgh (2000). This study was prospectively registered on May 7th, 2021 in the German Clinical Trial Register (reg. number: DRKS00024435). All subjects gave their informed consent.

2.2. Experimental Set-up

The participants were seating in a chair equipped with an arm-rest with their eyes open. The vigilance of the participants was constantly controlled by the examiner during the experiment by asking them standard questions. Additionally, they were required to fill out a questionnaire, where they were also asked whether they felt tired or sleepy before, during and after the performance of the experiment.

TMS: Frameless stereotactically neuronavigated (Brainsight TMS Navigation, Rogue Resolutions Ltd coupled with a “Polaris-Vicra” infrared camera (NDI, Waterloo, Canada)) TMS (MagPro X100, MagVenture, Dantec S.A., Skovlunde Denmark) with a slightly angulated figure-eight coil (MC-B70) was applied in this study. This TMS device allows setting the triggering pulses and

thus choosing the pulse direction [posterior to anterior (Pa) vs. anterior to posterior (Ap)] as well as the pulse-configuration [Monophasic (Mo) vs. Biphasic (Bi)]. Note, in the case of the biphasic pulses the induced current direction in the brain is opposite to the initial current flow in the coil (Siebner et al. ‘Transcranial magnetic stimulation of the brain: What is stimulated?’ in preparation). Hence, in terms of clarity, the current direction of the second biologically relevant phase of the Bi-pulses is used to indicate the current direction throughout this manuscript, Fig. 1.

Recording System: Motor-evoked potentials (MEPs) were recorded via surface custom-built silver-chloride electrodes with 5 mm diameter of the silver surface. Electrodes were placed up to 2 cm away from each other (electrode impedance = 100 Ω – 10 k Ω , low-Pass filter = 2.5 kHz, sampling-frequency = 5 kHz). For amplification an isolated patient amplifier system (Digitimer, Ltd., Model D360, Garden City, UK) and an analogue–digital-converter (CED 3001, MICRA 1401 mk II; Cambridge Electronic Design Limited, Cambridge, UK) were used. “Signal” software (CED 1401, Cambridge Electronic Design, Cambridge, UK) was used for EMG recording.

Electrode Placement: After ultrasound determination of the diaphragmatic excursion, the active electrode (supposed to record the diaphragmatic MEPs) was placed either in the 7th, 8th or 9th intercostal space (ICS) on the right hand side (contralateral to stimulation) between the costochondral junction and the midclavicular line (MCL). The reference electrode was placed slightly laterally on the above rib, as recommended in a previous study (Demoule et al., 2003). An additional pair of electrodes was placed in the right 7th, 8th or 9th ICS ipsilateral to stimulation.

To control for signal contamination from activation of neighboring muscles, EMG recordings were obtained at two additional locations. One pair of electrodes was placed in the 6th ICR at the posterior axillary line (PAL) of the right side in order to record activity from the *M. latissimus dorsi* and *M. serratus anterior* (see (Demoule et al., 2003)) and a second pair of electrodes was placed above the abdominal wall at the MCL, approximately 4 cm below the rib cage registering the activity of the abdominal muscles (*M. rectus* and *M. transversus abdominis*).

Ultrasound: Ultrasound measures in B- and M-Mode were performed using SonoSite Edge[®] Ultrasound Machine (Fujifilm, Sono-site, Inc.) to verify the diaphragmatic activation during both TMS and electrical phrenic stimulation. A curvilinear low-frequency phased-array transducer-probe (2.5–5 MHz) was used to register the pattern-deviations of the right hemidiaphragm-excursion (contralateral to TMS, Fig. 2). The probe was positioned subcostally at the MCL and angled up- and rightwards, so that the ultrasound beam was projected perpendicular to the right dome of the diaphragm (Boussuges et al., 2009).

2.3. Experimental protocol

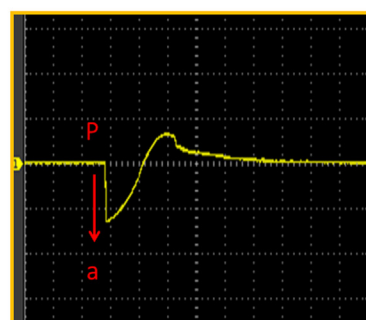
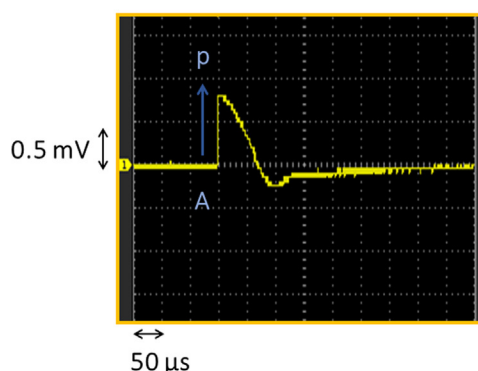
After the electrical phrenic nerve stimulation was completed, the experimental protocol for TMS with Mo-Pa, Mo-Ap, Bi-Pa and Bi-Ap was performed. Each of the four conditions was delivered in a separate stimulation-session with the order of the four sessions being pseudorandomized and separated by breaks of 5 to 10 minutes. A random list-creator was used, so that if a certain presentation combination was “picked” from the algorithm (e.g., 2 = Mo-Ap, 3 = Bi-Pa, 4 = Bi-Ap, 1 = Mo-Pa) for a concrete participant, this combination (e.g., 2.3.4.1.) was removed from the list in order not to be selected again. Each of the four stimulation-sessions started with determination of the hotspot and RMT, before the actual stimulation was performed.

Electrical Phrenic Nerve Stimulation: To control the electrode position and estimate the central motor conduction time (CMCT), electrical stimulation of the phrenic nerve was performed (Dig-

Current induced by MagPro X100 Measured by a Probe-Coil

Initial Current Direction (TMS-Coil): Mo-Ap
 Current Flow in the Brain: Mo-Ap
 Device-Nomination MagPro X100: “Reverse”

Mo-Pa
 Mo-Pa
 “Normal”



Initial Current Direction (TMS-Coil): Bi-Pa
 Decisive Second Component (Brain): Bi-Ap
 Device-Nomination MagPro X100: “Normal”

Bi-Ap
 Bi-Pa
 “Reverse”

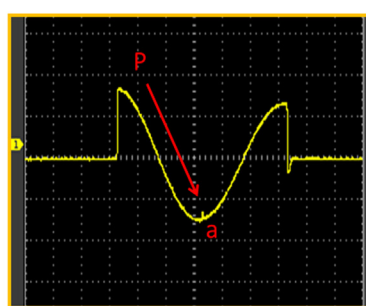
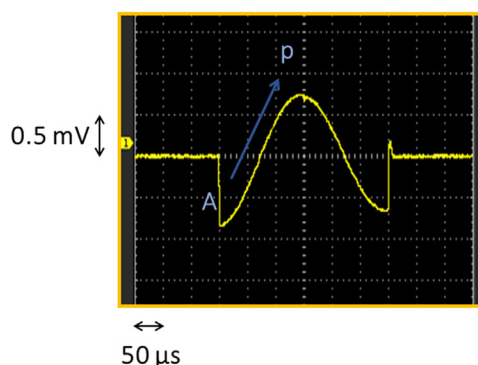


Fig. 1. Currents Induced in a Probe Coil. Transcranial magnetic stimulation (TMS)-pulse configurations of the present study tested using 1 cm-diameter probe coil i.e., the two different pulse configurations (biphasic pulses denoted as Bi vs. monophasic pulses denoted as Mo) and the two different current flow directions (posterior to anterior denoted as Pa vs. anterior to posterior denoted as Ap). The currents were executed by MagPro X100 Stimulator (MagVenture, Dantec S.A., Skovlunde Denmark) equipped with Dantec MCB70 coil. The recordings were made and stored by an oscilloscope (LeCroy 9350C, Chestnut Ridge, NY, USA). The graphs denote Volt (V) per microsecond (μ s). The stimulus intensity was 25% of the MSO for the biphasic and 27% of the MSO for the monophasic pulses. The stimulus-induced current direction in the brain is indicated with blue arrow for the Ap-oriented pulses and red arrow for the Pa-oriented ones. Note, that the second phase of the biphasic pulses, that is the physiologically relevant for inducing current flow into the brain is wider and opposite-oriented to the initial current direction in the coil. As shown on the plots the monophasic pulses produced by this device are not ideally monophasic but rather half-sine. Furthermore, the currents induced in the probe coil show asymmetrical waveform-components with different length. Thereby, the second effective phase of the biphasic stimuli comprises a length of approx. 200 μ s, because of the sharp rising slopes of the initial component, while the effective (first) phase of the monophasic stimuli comprises a length of about 100 μ s. Hence the physiologically effective second phase of the biphasic-pulses is twice as wider as the corresponding phase of the monophasic-pulses.

itimer MultiPulse®, model D185-Mark IIa, delivering square-wave pulses of 0.1-ms duration) by using bipolar electrodes placed 2 cm apart according to already validated protocol (Verin et al., 2002). The effectiveness of the stimulation was judged from the deviation of the real time pattern of the breathing movements registered via ultrasound in B- and M-mode and the absence of

concomitant coactivation of the cervical plexus and the long thoracic nerve (innervating the serratus anterior muscle).

Determination of the Motor Hotspots: The motor hotspot of the diaphragm was determined in a relaxed state (i.e., at the end of passive expiration). A multistep mapping procedure was applied for cortical mapping. In brief, the Cz, Fz and C3 points were

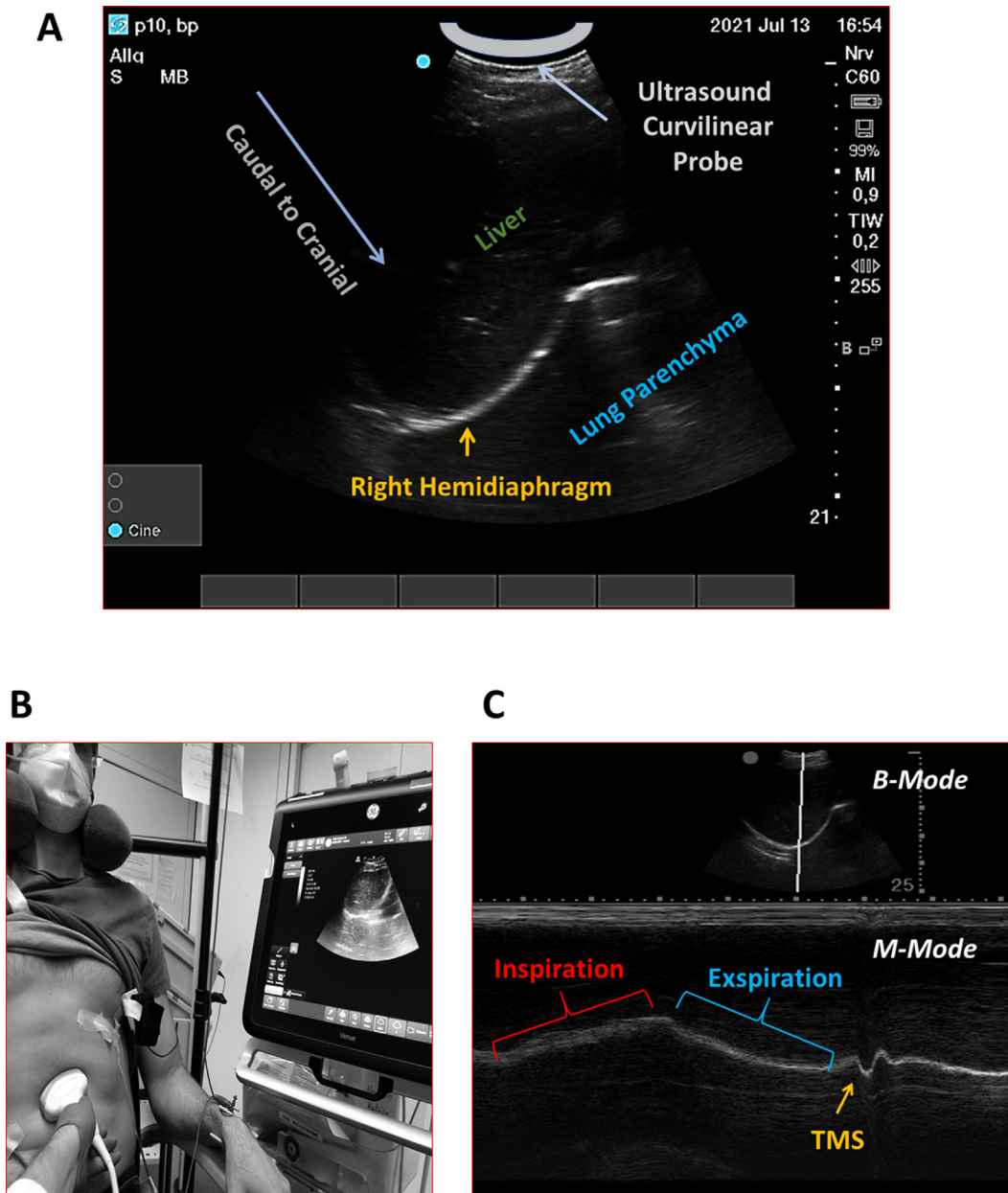


Fig. 2. Ultrasonography of the Diaphragm. **A.** Original B-Mode image showing the diaphragm of one test subject. **B.** While one investigator is performing the ultrasound for monitoring of respiratory movements, the other investigator is executing the TMS at end-expiration. **C.** A representative image showing a deviation of the normal pattern of diaphragm movements in M-mode after the transcranial magnetic stimulation execution with biphasic pulses (Bi-TMS).

denoted on the subject's head using the international 10–20 EEG placement system and then digitized onto a virtual MNI-head model. Then a 9-cells-virtual grid was placed over these points and a neuronavigated-TMS search was performed for the 9 cells (see Fig. 3 and Supplementary Fig. 1). The hotspot-search was performed separately for the four pulse-configurations (i.e., Mo-P, Mo-A, Bi-A and Bi-P).

Determination of the Motor Threshold and Stimulations: RMT was individually estimated for each condition (Mo-Pa, Mo-Ap, Bi-Pa, Bi-Ap) by using a freely available software package “Transcranial Magnetic Stimulation Motor Threshold Assessment Tool” by Friedemann Awiszus & Jeffrey J. Borckardt that relies on adaptive threshold-hunting (Borckardt et al., 2006) based on maximum-likelihood parameter estimation by sequential testing (ML-PEST) (Awiszus, 2003; Awiszus et al., 1999; Groppa et al., 2012). Note, that the initial stimulation-intensity for determination of the hot-

spot was chosen to be lower for the biphasic-TMS (75% MSO) than for the monophasic-TMS (90% MSO) based on research using FDI-protocols (Sommer et al., 2006; Stephani et al., 2016) and own pilot testing with the members of our laboratory. However, once the hotspot was found, the stimulation-intensity was gradually decreased in steps of 5% until it evoked no more diMEPs. Then the lowest possible intensity that still evoked diMEPs was entered into the ML-PEST. The ML-PEST assessment was performed until a TMS-intensity that produced diMEPs with peak-to-peak amplitudes $\geq 50 \mu\text{V}$ was found to have a confidence interval $> 95\%$. Once the RMT was determined, the stimulation was performed at 120%-RMT. 12 to 17 pulses were executed for each condition. If the determined RMT-value exceeded MSO, the stimulation was performed with 100%-MSO. After artifact rejection was accomplished (Supplementary Fig. 2) the MEPs were further statistically analyzed.

Cortical Stimulation Mapping

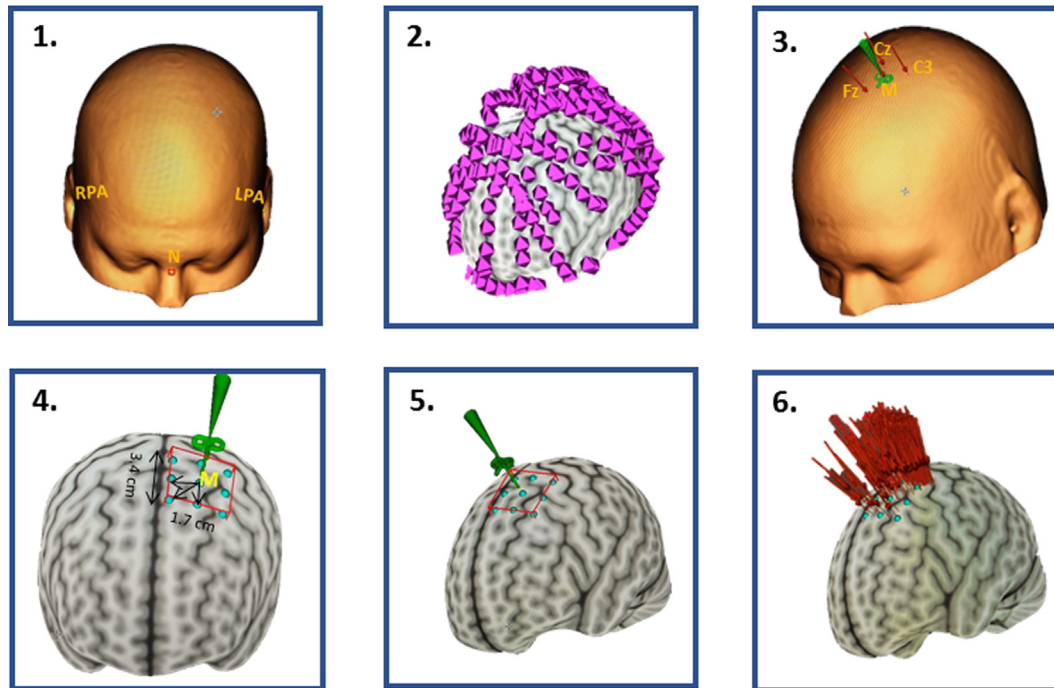


Fig. 3. Multistep Neuronavigated Cortical Mapping. 1. Defining the nasion (N) and the left (LPA) and right preauricular areas (RPA) on the test subject. 2. Scaling through picking the most frontal, temporal, parietal and occipital points from the subject's head-topography and overlying them with the MNI-head model. 3. Defining the Cz, Fz and C3 points according to the international 10–20 EEG placement system, marking them with a colored pencil on participants head. Defining the midpoint (M–point) on the distance between C3 and Fz on the head of the participant's head, so that “M” roughly corresponds to the average diaphragm stimulation coordinates (1 to 3 cm lateral of the midline and about 1 to 2 cm anterior to the auricular plane). Digitizing the Cz–, C3–, Fz– and M–Coordinates from the subject's head onto the established three-dimensional individual head model. 4., 5. Placing a digital 3x3-points virtual grid with a center corresponding to the M–point onto the head model. 6. Starting neuronavigated-transcranial magnetic stimulation (TMS) stimulation over the nine points of the grid in pseudorandomized order. The coil is being manually hold tangentially to the head and over each point of the grid and rotated arbitrarily in increments of 90° until the best corresponding diMEP-amplitudes are being found. Once the hot-spot is being determined for a certain pulse-configuration, the stimulation with the same wave but a reverse current direction is being firstly performed over the already defined hotspot. However, the search for the hotspot is being further carried on until the examiner refereed that there was no other hotspot or coil orientation that provided larger diMEPs-amplitudes. After the best coil orientation is found rotating the coil in 90° increments, the coil is being additionally rotated clockwise and anticlockwise or vice versa over this point in increments of 25° until the best MEP-amplitudes are found.

2.4. Data analysis and statistics

The major goal of this study was to probe the effect of four different TMS pulse-configurations (i.e., Mo-Ap vs Mo-Pa vs Bi-Ap vs. Bi-Pa) on the motor-evoked potentials of the diaphragm. Therefore, a 2×2 -repeated-measures (RM) ANOVA with factors “Waveform” and “Current Direction” was applied to examine the effects of the four pulse-configurations. Hence, the values of the RMT, Latency and Amplitude of the diMEPs represent the dependent variables, whereas the stimulation parameters: Waveform (Monophasic (Mo) vs. Biphasic (Bi)) and the Current Direction (Anterior-posterior (Ap) vs. Posterior-anterior (Pa)) represent the independent variable. Thereby, three separated RM-ANOVAs were performed for RMT, Latency and Amplitude. *Post-hoc t*-tests were paired and two-tailed. *Bonferroni correction* was applied for all statistical analyses.

For data collection and preprocessing of the variables MEP-Latency (time difference between stimulus artefact and MEP/CMAP onset) and MEP-Amplitude (voltage difference between positive and negative peaks) previously detailed procedures were used (Sommer et al., 2006; Stephani et al., 2016). R-Studio (R-Studio, PBC, Boston, MA) was used for data analysis, statistics and creating the figures and plots.

2.5. Descriptive findings

No side effects were reported after ending the TMS stimulation. However, one subject could not complete the experiment due to a reflexive syncope that occurred directly after electric phrenic stimulation. Hence, 19 subjects were stimulated using TMS.

2.6. Motor hotspot

The initial idea of the present study was first to determine the best diaphragmatic stimulation site over the left hemisphere and then to keep the TMS coil constant over these stimulation coordinates in order to probe the effect of four different pulse-configuration (Mo-Ap, Mo-Pa, Bi-Ap and Bi-Pa) on the diMEPs, as previously done for the hand muscles (Antal et al., 2002; Sommer et al., 2006; Stephani et al., 2016). However, a preliminary study with six members of our team showed that after the best stimulation site for a certain condition was found (e.g. for Bi-Ap), either very low MEP-amplitudes or not any were evoked, when the pulse-configuration was altered (e.g., from Bi-Ap to Mo-Pa) even if exactly the same stimulations' coordinates were used. Thus, a neuronavigated-TMS search for the best stimulation coordinates was performed for every condition separately. Additionally, the

coil orientation had to be adjusted from 0 to 360 degree until the lowest effective stimulation-intensity for each pulse-configuration was assessed (Supplementary Table 1). Despite that, 6 individuals (from 19) still did not respond to the monophasic-TMS. In contrast, the biphasic-TMS elicited diMEPs in all 19 subjects (only one subject did not respond to Bi-Pa-TMS). Furthermore, the biphasic-TMS required less time and number of single-pulses for determination of the motor hotspot as the monophasic-TMS (Supplementary Fig. 3).

Thereby, the present results revealed similar spatial distribution of the hotspots for TMS with Mo-Ap-, Mo-Pa, Bi-Pa and Bi-Ap-stimuli, Fig. 4. The mean stimulation coordinates for the different pulses were as follow; Mo-Pa = 1.94 cm (from 0 cm to 4 cm) later to midline (Cz-Fz) and 0.34 cm (from -2 to 2 cm) anterior/posterior to auricular plane (Cz-C3); Mo-Ap = 1.9 cm (0 to 3.5 cm) and 0.25 cm (-1 to 2 cm); Bi-Ap = 1.8 cm (0 to 4 cm) and 0.12 cm (-2 to 2 cm); Bi-Pa = 1.92 (1 to 4 cm) and 0.24 cm (-1 to 2 cm).

Although the different stimulation angels (i.e. coil orientation from 0 to 360 degree) showed some preferences regarding the four conditions, no specific pulse-configuration dependent stimulation-angle was found, Fig. 5. Even when it was expected that by reversing the current direction (e.g., changing the Mo-Ap to Mo-Pa) the best diMEP-amplitudes would be evoked by turning the coil to 180 degree (opposite to the previous current flow) this was not the case in most of the subjects.

3. Results

3.1. Resting motor threshold

The mean RMT of the two biphasic pulse-configurations (Bi-Ap = $69.21 \pm 17.95\%$, Bi-Pa = $68.05 \pm 16.59\%$) was lower than the RMT of the two monophasic pulses (Mo-Pa = $87.42 \pm 13.92\%$, Mo-Ap = $88.15 \pm 12.96\%$), Fig. 6. RM ANOVA revealed a significant main effect for Waveform ($F(1,18) = 126.5$, $p < .001$) confirmed by *post hoc* pairwise comparisons ($p < .001$ for comparison between Bi-Ap and Mo-Ap, and $p < .05$ for all other comparisons between monophasic and biphasic pulses).

3.2. Diaphragmatic Motor-evoked potentials

In all cases where diaphragmatic motor responses could be evoked, the stimulation was accompanied by concomitant ultrasound registration of diaphragm movements in M- and B-Mode (Fig. 2).

MEP-amplitude: The biphasic-TMS yielded generally greater diMEP-amplitudes (Bi-Ap = 0.287 ± 0.219 mV; Bi-Pa = 0.268 ± 0.211 mV) than the monophasic-TMS (Mo-Pa = 0.194 ± 0.176 mV; Mo-Ap = 0.189 ± 0.124 mV) (significant main effect Waveform ($F(1,18) = 4.605$, $p < .05$)), $p < 0.05$ for all *post-hoc* comparisons between biphasic- and monophasic-TMS, Fig. 7. Additionally, scatter plots with quadratic regression analysis were generated that

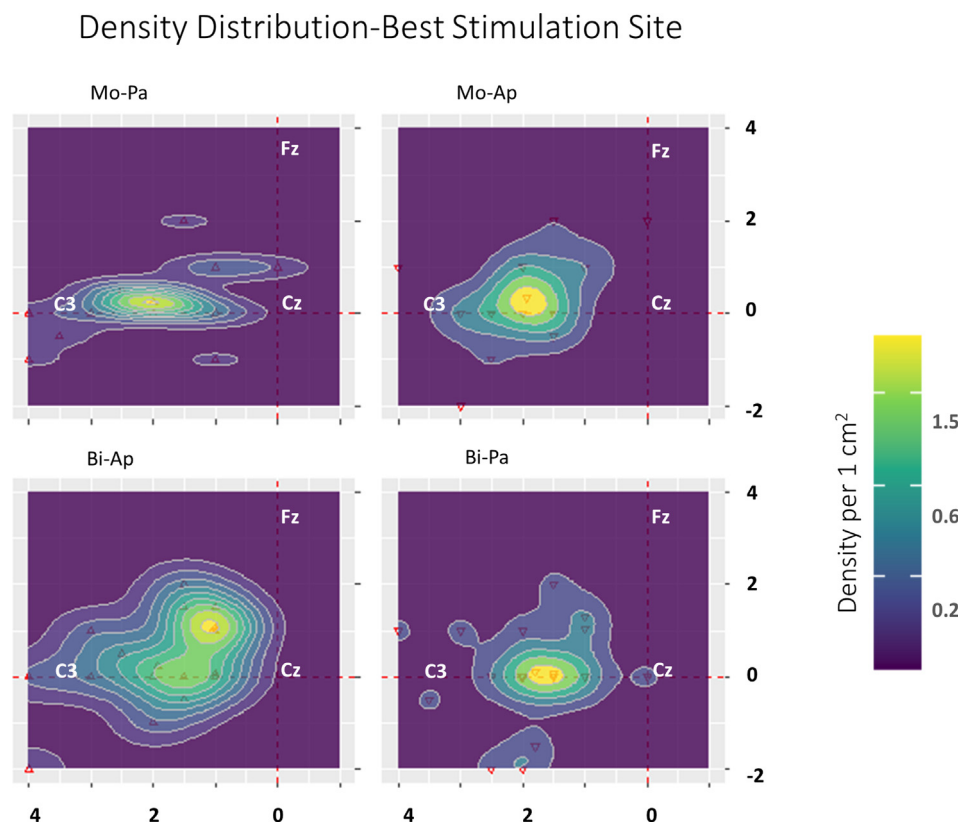


Fig. 4. Motor Hotspots Density Distribution Plots. The plots show the spatial density distribution of the best stimulation site (motor hotspot) across the 19 subjects and four transcranial magnetic stimulation pulse-configurations (TMS) used in this study, viz. biphasic pulses denoted as Bi vs. monophasic pulses denoted as Mo executed as two different current flow directions (posterior to anterior denoted as Pa vs. anterior to posterior denoted as Ap). The density distribution of the best stimulation sites is plotted according to individual subjects' coordinates Cz-Fz (lateral to Midline) and Cz-C3 (anterior or posterior to Auricular Plane) over the left hemisphere. The triangles indicate the exact hotspot coordinates for each of the 19 subjects. The initial current flow in the coil (opposite to the second decisive phase in the brain in the case of the Bi-TMS) is indicated with triangles pointing up for "Pa"- and triangles pointing down for "Ap"-current flow. The scale indicates subjects' hotspots density distribution per 1 cm². The density plots show similar patterns of spatial distribution of the motor hotspots for TMS with Mo-Ap, Mo-Pa, Bi-Pa and Bi-Ap-stimuli (i.e. 1 to 4 cm lateral of the midline and about 1 to 2 cm anterior to the auricular plane).

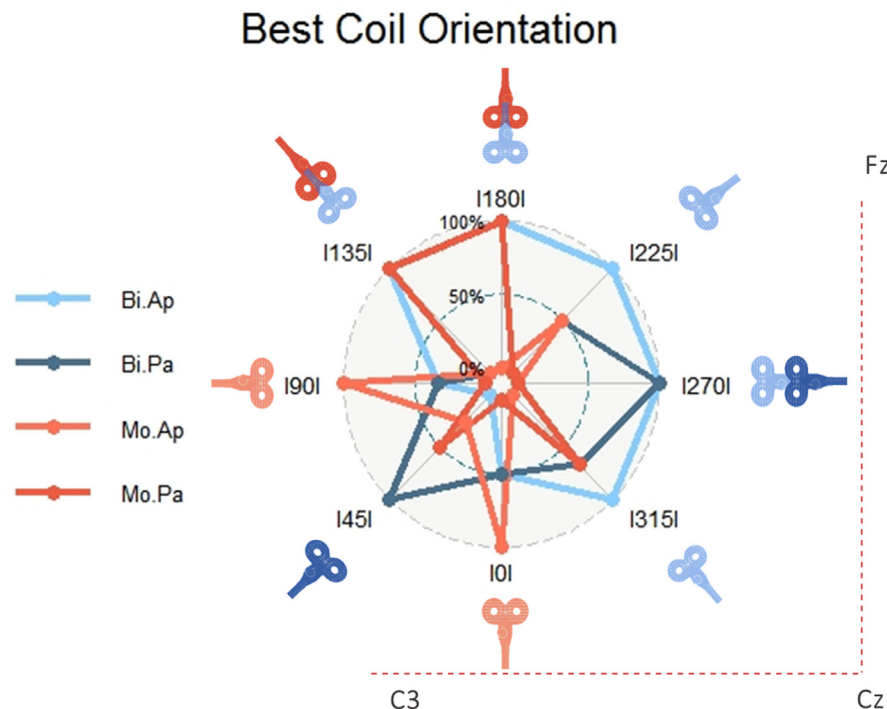


Fig. 5. Radar Map showing stimulation-angle preferences for each of the four pulse-configurations in percentage. The two different transcranial magnetic stimulation (TMS)-pulses (biphasic pulses denoted as Bi vs. monophasic pulses denoted as Mo) are color-coded; red for monophasic- and blue for biphasic-TMS. Bi- and Mo-pulses were executed either with posterior-anterior directed current flow (denoted as Mo-Pa and Bi-Pa) or as anterior-posterior directed currents (denoted as Mo-Ap and Bi-Ap). The preferences of the different angles (coil orientation from 0 to 360 degree) for a certain TMS-simulation condition are displayed on the radar map as percentage of time a certain condition yielded the best responses across the subjects at this concrete angle. The vertical axis of the radar map indicates percentage. In case of TMS-coil orientation at angles 0- and 90-degree- Mo-Ap pulses are best-suited (100% of the cases), from 135 to 180 degree Mo-Pa and Bi-Ap pulses are suitable and by coil orientation from 180 to 315 degree also Bi-Ap-TMS yielded the best diMEP-responses. If the TMS-coil is placed at 45- and at 270-degree, Bi-Pa pulses produced the best diMEPs in 100% percent of the cases, and in about 50% and 70% of the cases at 0- and at 315-degree, respectively.

show a negative relationship between the RMTs and diMEP-amplitudes, i.e. the higher the calculated RMT values the lower the amplitude of the motor responses after 120 %RMT-TMS, see Fig. 8. The negative relationship between RMT and diMEP-amplitude appeared to be linear by the Bi-Ap stimulation and tended to be linear in case of Mo-Ap and Mo-PA pulses. Moreover, the individual motor responses showed higher amplitudes after biphasic-TMS than the after monophasic-TMS, although smaller RMTs and respectively lower stimulation intensities (120%-RMT) were determined for the Bi-pulses.

MEP-latency: Considering their biological effects, the anterior-posterior (Ap) oriented Mo- and Bi-pulses yielded diMEP with shorter latencies than the posterior-anterior oriented ones, Fig. 7. The averaged onset latencies were as follow: Bi-Ap (16.35 ± 1.35 ms); Bi-Pa (16.69 ± 1.91); Mo-Ap- (16.71 ± 1.76 ms); Mo-Pa (17.47 ± 1.76 ms). RM ANOVA revealed a significant main effect for Current Direction [$F(1,18) = 6.263, p < .05$]. The following *post hoc* comparison showed that diMEP-latencies after TMS with Bi-Ap and Mo-Ap pulses were significantly shorter than the diMEPs after stimulation with Mo-Pa pulses ($p < .05$).

Additionally, bilateral diMEPs were evoked in five subjects after biphasic- and in only one case after monophasic-TMS. In those cases, the ipsilateral diaphragmatic MEPs appeared later than the contralateral ones (Supplementary Fig. 4).

Central Motor Conduction Time (CMCT): In all 19 subjects the electrical stimulation of the right phrenic nerve elicited ipsilateral diaphragmatic compound motor-evoked potentials (CMAP) (mean onset-latency 5.15 ± 1.31 ms). Expectedly, the central motor conduction times (CMCTs) of the four pulse-configurations resembled the pattern of diMEP-latencies (Bi-Ap = 11.19 ± 1.85 ms; Bi-Pa = 11.53 ± 1.91 ms; Mo-Ap = 11.56 ± 1.77 ; Mo-Pa = 12.32 ± 1.76 ms).

Paired simple *t*-tests was used here to reveal that the CMCT after TMS with Mo-Ap is significantly shorter than after TMS with Mo-Pa pulses ($t = 2.1632, df = 18, p = 0.0442$). Additionally, the TMS with Bi-Ap pulses was followed by significantly reduced CMCTs as compared to TMS with Mo-Pa pulses ($t = 2.805, df = 18, p = 0.0117$).

3.3. Coactivation of extradiaphragmatic muscles

Coactivation of extradiaphragmatic muscles was observed in the right 6th ICR, PAL (M. latissimus dorsi and serratus anterior) in 10 and 11 and in the abdominal electrodes in 11 and 8 subjects when using Bi-Pa and Bi-Ap pulses, respectively. Correspondingly, coactivation occurred in 11 and 8 as well as in 6 of these subjects after TMS with Mo-Pa and Mo-Ap pulses, respectively (Supplementary Fig. 5). In accordance with previous research the motor activation of the extradiaphragmatic muscles always appeared in different latency interval as compared to the diaphragmatic motor-evoked potentials (Demoule et al., 2003). Hence, the mean MEPs of the serratus/latissimus muscles appeared earlier than the diMEPs i.e., mean latencies: $13.09 \text{ ms} \pm 2.22$ and $12.93 \text{ ms} \pm 2.42$ for the Bi-Ap and Bi-Pa and $12.97 \text{ ms} \pm 1.25$ and $12.79 \text{ ms} \pm 2.53$ for the Mo-Ap and Mo-Pa, whereas the activation of the abdominal muscles appeared later than the diMEPs i.e., mean latencies: Bi-Ap $p = 19.89 \text{ ms} \pm 3.19$, Bi-Pa = $22.04 \text{ ms} \pm 3.38$, Mo-Ap = $20.22 \text{ ms} \pm 2.97$, Mo-Pa = $20.55 \text{ ms} \pm 2.70$.

4. Discussion

The present study demonstrated that biphasic as compared to monophasic-TMS pulses require significantly less stimulation

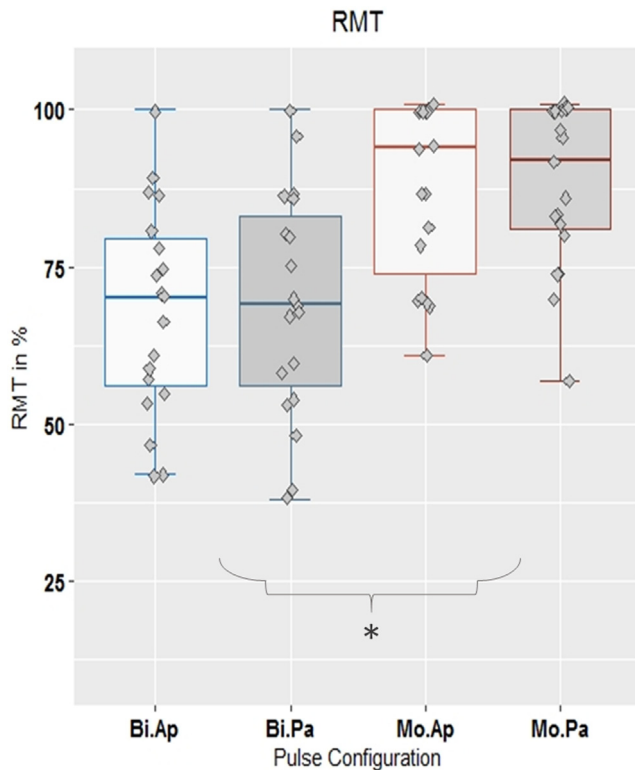


Fig. 6. Differences in the Resting Motor Threshold (RMT) Boxplots represent the RMTs for four transcranial magnetic stimulation (TMS)-conditions: monophasic (Mo) with posterior-anterior current flow (Pa) and monophasic (Mo) with anterior-posterior current flow (Ap) pulses, as well as biphasic-pulses (Bi) with either posterior-anterior (Pa) or anterior-posterior currents (Ap). The ordinate shows percent of maximal stimulator output. The individual RMTs among the subjects are indicated with grey rhombs. The lines that divide the boxes into two parts represent the median. Black asterisk indicates significant difference between the conditions.

intensity to induce diMEPs. These findings are similar to the earlier data that probed the efficiency of the TMS parameters to elicit MEP from the hand-muscles (Kammer et al., 2001; Niehaus et al., 2000; Sommer et al., 2006; Stephani et al., 2016) as well as the outcome of rTMS studies that showed that the monophasic-rTMS is less effective than the biphasic-rTMS (Antal et al., 2002; Sommer et al., 2002). Moreover, our data demonstrated significantly higher amplitudes of the diaphragm motor responses to biphasic- than to monophasic-pulses. This behavior may be explained by the different biophysical properties of the two waveforms. The monophasic and the biphasic pulses are the two most widely used TMS waveform-configurations (e.g., review (Rossini et al., 2015)). Monophasic pulses induce a single positive current in the brain (Rossini et al., 2015). Biphasic pulses on the other hand have negative and positive phases and consequently induce at least two currents of inverse direction with a wider and longer reversed-second phase than the initial rising phase. This leads to a complex interaction between the produced current and the neural tissue which makes the biphasic pulses more effective in producing current fluxes in the brain (Kammer et al., 2001; Rossini et al., 2015).

However, it should be noted here, that the present findings regarding RMT could not entirely be assigned to “phase-differences” between the monophasic and biphasic pulses because the two effective waveform phases are also different in pulse duration (Sommer et al., 2006; Stephani et al., 2016).

It has been suggested that wider pulses are more efficient in inducing excitation (Halawa et al., 2021). Moreover, the monophasic pulse produced by the present TMS-device (i.e. MagProX100) is more asymmetrical than the physiologically relevant second wider

phase of the biphasic pulse. Additionally, the intensity of given stimulus indicated by the device is supposed to be derived from the peak current in the initial 50 ms (peak intensity) of the stimulus waveform (Sommer et al., 2006) but not the overall intensity of whole waveform. Therefore, the wider second phase of the biphasic- compared to the monophasic-pulses might also explain one of the findings of the present study i.e., that biphasic-TMS executed with lower stimulation intensity yielded greater MEP-amplitudes than the monophasic-TMS. Future studies should also explore the phase-dependent efficiency of activating the diaphragm with more emphasis on the strength-duration relationship (Halawa et al., 2021; Peterchev et al., 2013; Shirota et al., 2016) by means of controllable pulse-parameter devices such as cTMS. Such devices provide enhanced flexibility, such as adjusting the phase and duration of the different TMS-waveform components (Peterchev et al., 2013) that can contribute to novel insights in the neurophysiology of the cortical activation of the breathing muscles.

The circumstance that monophasic pulses require higher stimulation-intensity than biphasic pulses represents a considerable limiting factor in the inspection of the diaphragm by TMS. In fact, most of such studies, with some exceptions (e.g., (Laviolette et al., 2013; Miscio et al., 2006)), have been performed by means of devices providing only monophasic pulses (i.e. Magstim 200), hence the stimulations have been accomplished with maximal or supramaximal TMS-device output (e.g., see (Davey et al., 1996; Kawaguchi et al., 2000; Lefaucheur and Lofaso, 2002; Lissens and Vanderstraeten, 1996; Maskill et al., 1991; Murphy et al., 1990; Rosler et al., 1994; Welch et al., 2020; Zifko et al., 1996)). In the present study we were not able to derive any MEPs of the diaphragm in 6 out of 19 subjects when using monophasic-TMS even at 100% MSO, whereas diMEPs were derived in all 20-test subjects after stimulation with biphasic-posterior-anterior-TMS. In a previous TMS research, diMEPs could also not be evoked in some individuals using a figure-of-eight coil and applying monophasic-TMS even at maximal intensity. Demoule et al., 2003 for example, obtained MEPs from the diaphragm in seven out of nine subjects by using focal monophasic-TMS with a figure-of-eight coil, whereas diMEPs were elicited in all nine subjects when applying non-focal TMS over the vertex using a circular coil (Demoule et al., 2003). In this context, the technique of “voluntary contraction of the muscle”, namely stimulation in voluntary inspiration and/or “reflex drive to ventilation” with inhaled CO₂ were utilized in several previous experiments to lower the RMT and thus to facilitate the generation of diMEPs (Davey et al., 1996; Kawaguchi et al., 2000; Sharshar et al., 2003; Similowski et al., 1996a; Verin et al., 2004). Indeed, stimulation in active inspiration may provide additional practically relevant clinical information about the efficacy to activate the diaphragm in pathological state. However, unlike other skeletal muscles, the diaphragm contractions during normal respiration are automatically and involuntary driven by the respiratory centers in the brainstem. The same degree of muscle contraction should be maintained, in order to accurately calculate the active motor threshold (AMT) in order to compare the effects of different TMS-pulse configurations. One option to control that is to ask the test subject to breath voluntary in order to stimulate at maximum contraction (using twitch pressure technique, see (Similowski et al., 1996b)) and/or when the esophageal pressure begins to fall in early inspiration (e.g., see (Sharshar et al., 2003; Straus et al., 2004; Verin et al., 2004)). Nevertheless, voluntary inspiration may result in stronger coactivation of extradiaphragmatic muscles and signal contamination of the surface recordings (Demoule et al., 2003). Therefore, in the present study TMS was performed at the end of passive expiration.

Another strategy that has been employed in TMS diaphragm research is the use of double-cone coils (Hopkinson et al., 2004;

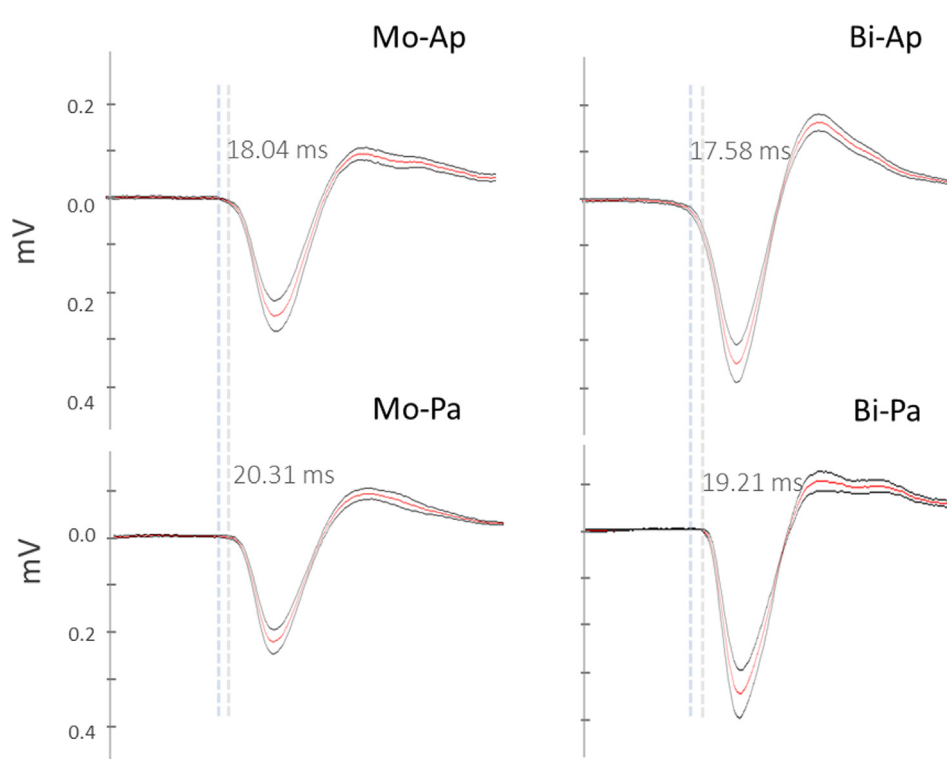


Fig. 7. Diaphragmatic MEPs of one representative Subject. The figure denotes the mean MEPs of the diaphragm averaged over 17 stimulation-frames after transcranial magnetic stimulation (TMS) with monophasic (Mo) and biphasic pulses (Bi), with anterior to posterior (Ap) and with posterior to anterior (Pa) oriented currents. The different pulses are denoted by means of their biological effects in the brain as: Mo-Ap, Bi-Ap, Mo-Pa, Bi-Pa (viz. opposite to the physical orientation of the initial current flow into the coil in case of Bi-TMS). The dispersion of the 17 diMEP-waveforms from the mean diMEP-waveform (red) is indicated with error bars (grey lines). The MEP-latencies onsets for the different pulse configuration are indicated with blue and grey dashed lines.

Sharshar et al., 2004a, 2004b; Verin et al., 2004). A double-cone coil may be used if the examiner is unable to obtain diMEPs, however this coil provides less focal stimulation and requires high stimulation intensities (Similowski et al., 1996d). On the other hand, broader stimulation with maximal intensity could facilitate the search for the best stimulation site of the diaphragm due to the very small cortical representation of this muscle, especially in the absence of a neuronavigational system. The earlier TMS studies that attempted to stimulate the diaphragm employed “non-focal stimulation” by means of Magstim 200 devices with a circular coil around the vertex (Lissens, 1994; Lissens and Vanderstraeten, 1996; Maskill et al., 1991; Murphy et al., 1990). Though, such a non-focal stimulation could also lead to uncertainty about the validity of the registered MEPs when using superficial electrodes, due to coactivation of muscles neighboring the diaphragm and respective signal contamination (Demoule et al., 2003). Stimulation with a figure-of-eight coil after performing a cortical mapping and RMT determination provides more focal stimulation (Rossini et al., 2015). In fact, very few TMS diaphragm studies estimated the motor threshold (e.g., (Welch et al., 2021, 2020)) using the R-R method (Rossini et al., 1994), or any other recommended algorithm in the practical guidelines for use of TMS (Groppa et al., 2012; Rossini et al., 2015). In the present study, neuronavigated-TMS with a figure-of-eight coil was used for determination of the best stimulation site of diaphragm over the dominant hemisphere for each pulse-configuration. Additionally, the RMT was precisely assessed for every subject and for each condition independently (Mo-Pa, Mo-Ap, Bi-Pa, Bi-Ap) based on maximum-likelihood parameter estimation by sequential testing (ML-PEST) (Awiszus, 2003; Awiszus et al., 1999). Neuronavigated-TMS allows small real-time adjustments of the TMS coil over the hotspot

(Ruohonen and Karhu, 2010), which finally enables more focal stimulation with respect to individual anatomy and lower stimulation intensity (Zmeykina et al., 2020).

Thereby, our results showed small intraindividual shifts regarding the best stimulation site. However, the mean stimulation coordinates showed no waveform-specific spatial distribution. The best stimulation site for all four conditions (Mo-Ap, Mo-Pa, Bi-Ap and Bi-Pa) was approximately 0 to 4 cm lateral to the midline and up to 2 cm anterior to the auricular plane, which is in line with previous research using figure-of-eight coil and monophasic-TMS (e.g., see (Demoule et al., 2003; Kawaguchi et al., 2000; Khedr and Trakhan, 2001; Zifko et al., 1996)). Studies with double-cone coils also reported the best stimulation site to be slightly lateral and anterior to the vertex (Hopkinson et al., 2012, 2004; Sharshar et al., 2004a; Verin et al., 2004). Unlike the stimulation coordinates, no clear angle-specific (coil orientation-specific) preference for the four pulse-configurations was found, and the best coil orientation varied among the conditions and the participants. This outcome could be explained by the fact that the coil orientation was adjusted from 0 to 360 degree until the lowest effective current flow was assessed and as a result the best coil orientation for stimuli with reverse current direction was not always at 180 degree to the opposite-oriented currents. In this context, a recent computational model study demonstrated that changing the stimulation waveforms or the direction of the currents ultimately determines the focus of maximal electromagnetic exposure in the targeting cortical structures (Abera et al., 2020). Therefore, the least movement or rotation of the coil alters the focus of maximal electromagnetic field exposure in the microstructures of the cortical layers. Thus, a precise optimization of the stimulation coordinates is required when

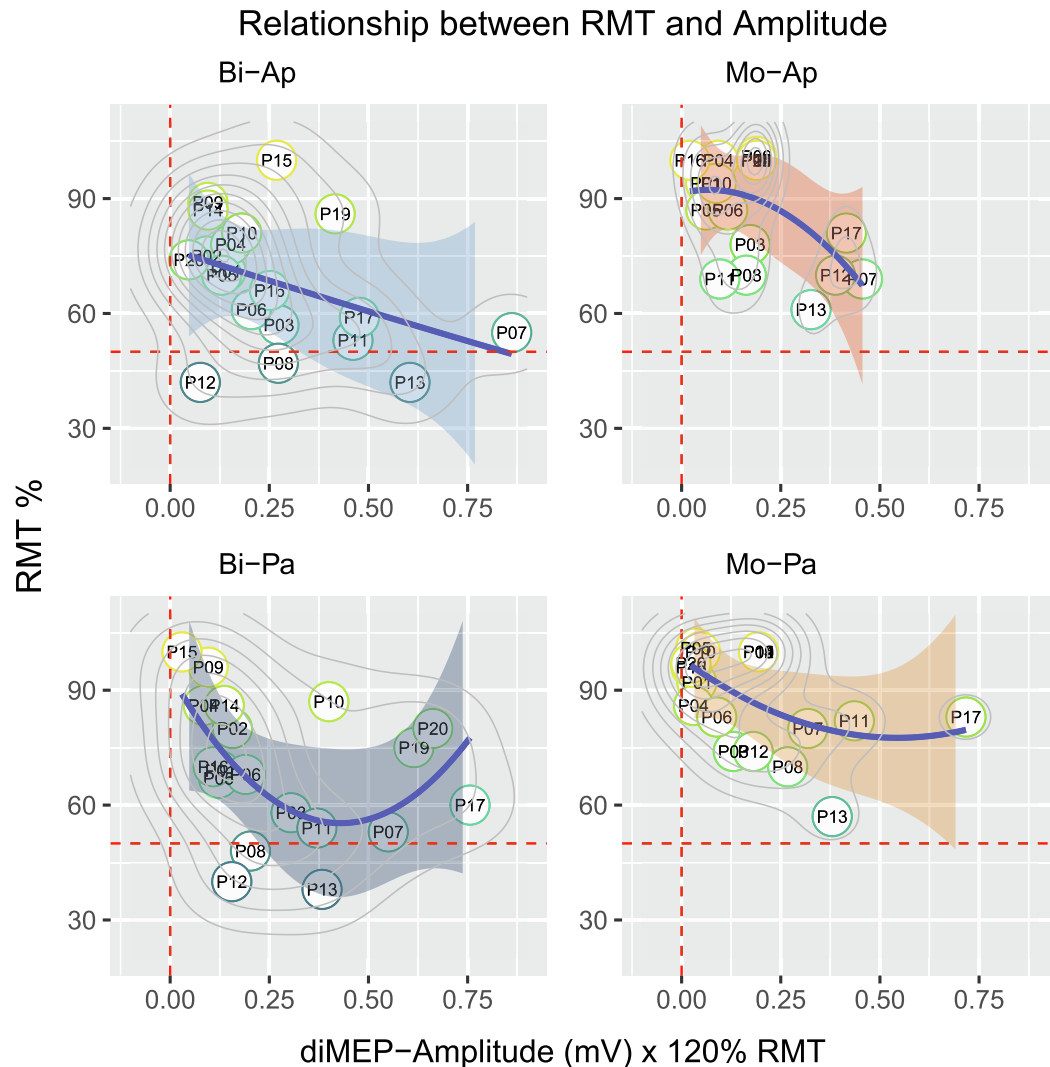


Fig. 8. Scatter Plots represent the Relationship between the Resting Motor Threshold (RMT) and diMEP-Amplitude for four TMS-Modalities. The relationship is modeled using quadratic regression. The biphasic- and monophasic-pulses are denoted as Bi and Mo respectively. The posterior-anterior directed currents are denoted as Pa (viz. Mo-Pa and Bi-Pa) and the anterior-posterior directed currents as Ap (viz. Mo-Ap and Bi-Ap). The x-axis displays diMEP-amplitude evoked after transcranial magnetic stimulation (TMS) with 120% above the individually calculated RMT-values, the y-axis displays RMT in %. The average individual responses for each condition are denoted as number of the concrete participant in the study (e.g., P19 for participant-number 19). The colorful circles around the participants indicate stimulation output, i.e. the higher the stimulation intensity the yellower the response, the lower the intensity the greener the circle. Regression lines are denoted with thick blue lines. Confidence intervals (>99%) for the four conditions are displayed with colorful bands around the regression lines (red for monophasic- and blue for biphasic-TMS). The plots show tendency of negative correlation between the RMTs and diMEP-amplitudes, i.e. the higher the calculated RMT-value the lower the diMEP-amplitude. This negative correlation is linear in the case of Bi-Ap and tends to be linear by Mo-Ap and Mo-Pa. Note also, that the motor-evoked responses showed higher amplitudes after biphasic-TMS than after monophasic-TMS, despite the smaller determined RMTs and the lower stimulation intensity, respectively.

changing the TMS-parameters if the target muscle has a very small cortical representation like the diaphragm.

Additionally, the present study revealed that biphasic- and monophasic-TMS with anterior-posterior oriented currents in the brain yielded diMEPs with shorter latencies than the TMS with posterior-anterior current flow. This implies that reversing the initial direction of the current flow in the TMS-device affects the latencies of the diMEP in a reverse manner, i.e. “reversed” Mo-pulses yielded shorter diMEP-latencies than the “normal” ones, whereas the effect was opposite regarding biphasic pulses (latencies to normal-Bi- were shorter than latencies to reverse-Bi-Pulses). Since the second quarter cycle of the biphasic waveform, that is oppositely oriented to the initial quarter cycle, is the physiologically relevant component of this pulse-configuration (Maccabee et al., 1998), the recruitment pattern of biphasic pulses with a first (initial) phase of posterior to anterior (Pa) and a second phase of anterior to posterior (Ap) direction is similar to that of

monophasic pulses of anterior-posterior (Mo-Ap) current direction and vice versa, the biological effects of the “reverse” Bi-pulses with Ap-initial phase correspond to those of the Mo-Pa “normal” pulses (Lazzaro et al., 2001). Lazzaro et al. (2001) suggested that the Ap- and the Pa-oriented stimuli activate different cortical structures and/or different sites of the same sets of interneurons that are possibly differentially sensitive to the current direction (Lazzaro et al., 2001). Therefore, the specific biological properties of the two main stimulation waveforms explain the specific effects regarding the MEP-latencies after stimulation with monophasic and biphasic pulses from the present study. On the other hand, the finding that Ap directed currents in the brain generate shorter-latency diMEPs does not correspond to the findings about hand muscles that demonstrated shorter-latency MEPs after Pa-TMS (Day et al., 1989; Sakai et al., 1997; Stephani et al., 2016). One can speculate that this is an epiphenomenon of the profounder and smaller localization of the diaphragm cortical representation (Khedr and

Trakhan, 2001) in contrast to e.g. hand muscles in M1 that are more superficially located and relatively tangential to the surface. In addition, anteriorly-directed induced current flow preferences were also reported for the leg-muscles motor area (viz. most effective stimulation with figure-of-eight coil and Magstim-200 by coil orientation between 0 and 135 degree) (Terao et al., 2000). Hence, the Ap and Pa directed currents may flow from different (opposite) angle through the cortical layers of the more deeply located structures like the diaphragm representation which leads to faster neuronal depolarization and/or stimulation of different subsets of interneurons. Furthermore, it has been demonstrated that the left and right hemidiaphragm are reciprocally represented and close to each other into the right and left motor cortex and unilateral TMS can activate as well contralateral as ipsilateral hemidiaphragm (Khedr et al. 2000).

Finally, we performed simultaneous surface recordings of the muscles neighboring the diaphragm, in order to probe the roll of the utilized TMS parameters in terms of diMEP signal contamination by those muscles most likely to be coactivated during diaphragmatic TMS. Indeed, in some cases TMS elicited MEPs at the surface recordings of latissimus dorsi and serratus anterior and the muscles of the abdominal wall. In line with Demoule et al., 2003 this activity was in a different latency-range as the diaphragmatic recordings. Specifically, in series of studies this research group validated the method of surface diaphragm-recordings by demonstrating that the latency of diaphragm-motor responses to both TMS and cervical phrenic nerve stimulation were not significantly different between simultaneous needle- and surface-electrode recordings (Demoule et al., 2003; Verin et al., 2002). However, the MEP-latencies obtained from the electrodes position supposed to register the diaphragm were significantly different from those of the extradiaphragmatic muscles (Demoule et al., 2003). Nevertheless, in our study the biphasic pulses more often caused coactivation of extradiaphragmatic muscles than the monophasic pulses. Moreover, in six cases the focal biphasic-TMS also evoked motor responses in the left hemidiaphragm (ipsilateral to stimulation), appearing one to several milliseconds later than the diMEP from the right hemidiaphragm (contralateral to TSM). We thus interpreted this outcome as an electrical signal irradiation through the diaphragm neural network from right to left hemidiaphragm after successful TMS activation.

Along with the diagnostic meaning, the diaphragmatic TMS has also therapeutic potential that should be utilized in the near future. The establishment of noninvasive techniques for fast identification and possible treatment of central and peripheral diaphragm dysfunction is appealing in times with increasing number of patients that develop respiratory failure and require mechanical ventilation. TMS may offer such a noninvasive, user friendly and economically beneficial therapeutic tool that does not require patients' volitional effort and hence can be conducted on ventilated patients in coma and/or general anesthesia. Future research should focus on the possibility to "condition" the breathing muscles using rTMSn of their cortical representation. For instance, it has been already shown by means of TMS and diaphragmatic MEPs that mechanical ventilation may lead to ventilator-induced changes in neural plasticity on a cortical level (Sharshar et al., 2004b). Hence, it is theoretically possible to produce changes in the excitability of the diaphragm cortical representation by means of rTMS. However, similar to our findings, these authors showed that the stimulus-modality may also affect the results of rTMS (Antal et al., 2002; Arai et al., 2007, 2005). On the one hand it has been demonstrated that monophasic repetitive stimulation has stronger after-effects on motor cortical excitability than biphasic rTMS (Arai et al., 2007), on the other hand a biphasic-TMS presented as train of low-intensity of high-frequency pulses was demonstrated to produce strong inhibition of the motor cortex (e.g., see (Arai et al.,

2009)). Regarding the current flow direction, it was found that posteriorly directed rTMS is best suited for induction of plasticity in moto cortex (Sommer et al., 2013). Therefore, future studies should probe the effects of the four most-widely used TMS-modalities from the present research (viz. Mo-Ap, Mo-Pa, Bi-Ap and Bi-Pa) by means of rTMS in order for the different phases of the respiratory process (inspiration and expiration) to be conditioned either in directions of inhibition or facilitation of breathing, which would have great therapeutic potential.

5. Conclusion

By comparing the motor responses of the diaphragm after monophasic and biphasic single-pulse TMS within the same group of subjects our study revealed that biphasic pulses are more effective in inducing motor responses in this muscle (Supplementary Table 2). Overall, the present research suggests that the choice of the TMS-stimulus modality may have a significant impact on the inspection of the corticospinal pathway to the diaphragm.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. IC was promoted by "German Research Foundation-Clinician Scientist" academic qualification program (DFG-CS) and received financial support to perform the experiments in parallel to his specialist training in the University Medical Center in Göttingen, project number: 413501650.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clinph.2022.03.019>.

References

- Abera AS, Wang B, Grill WM, Peterchev AV. Simulation of transcranial magnetic stimulation in head model with morphologically-realistic cortical neurons. *Brain Stimul* 2020;13(1):175–89.
- Antal A, Kincses TZ, Nitsche MA, Bartfai O, Demmer I, Sommer M, et al. Pulse configuration-dependent effects of repetitive transcranial magnetic stimulation on visual perception. *NeuroReport* 2002;13(17):2229–2223. <https://doi.org/10.1097/00001756-200212030-00013>.
- Arai N, Furubayashi T, Inomata-Terada S, Okabe S, Kobayashi-Iwata N, Hanajima R, et al. Effects of a high-frequency, low-intensity, biphasic conditioning train of TMS pulses on the human motor cortex. *Neurosci Lett* 2009;462(3):188–92.
- Arai N, Okabe S, Furubayashi T, Mochizuki H, Iwata NK, Hanajima R, et al. Differences in after-effect between monophasic and biphasic high-frequency rTMS of the human motor cortex. *Clin Neurophysiol* 2007;118(10):2227–33.
- Arai N, Okabe S, Furubayashi T, Terao Y, Yuasa K, Ugawa Y. Comparison between short train, monophasic and biphasic repetitive transcranial magnetic stimulation (rTMS) of the human motor cortex. *Clin Neurophysiol* 2005;116(3):605–13.
- Awiszus F. TMS and threshold hunting. *Suppl Clin Neurophysiol* 2003;56:13–23. [https://doi.org/10.1016/S1567-424X\(09\)70205-3](https://doi.org/10.1016/S1567-424X(09)70205-3).
- Awiszus F, Feistner H, Urbach D, Bostock H. Characterisation of paired-pulse transcranial magnetic stimulation conditions yielding intracortical inhibition or I-wave facilitation using a threshold-hunting paradigm. *Exp Brain Res* 1999;129:317–24. <https://doi.org/10.1007/s002210050901>.
- Borckardt JJ, Nahas Z, Koola J, George MS. Estimating resting motor thresholds in transcranial magnetic stimulation research and practice: A computer simulation evaluation of best methods. *J ECT* 2006;22:169–75. <https://doi.org/10.1097/01.yct.0000235923.52741.72>.
- Boussuges A, Gole Y, Blanc P. Diaphragmatic motion studied by M-mode ultrasonography. *Chest* 2009;135(2):391–400. <https://doi.org/10.1378/chest.08-1541>.
- Davey NJ, Murphy K, Maskill DW, Guz A, Ellaway PH. Site of facilitation of diaphragm EMG to corticospinal stimulation during inspiration. *Respir Physiol* 1996;106(2):127–35. [https://doi.org/10.1016/S0034-5687\(96\)00069-2](https://doi.org/10.1016/S0034-5687(96)00069-2).
- Day BL, Dressler D, Maertens de Noordhout A, Marsden CD, Nakashima K, Rothwell JC, et al. Electric and magnetic stimulation of human motor cortex: surface EMG and single motor unit responses. *J Physiol* 1989;412(1):449–73.

- Dayer MJ, Jonville S, Chatwin M, Swallow EB, Porcher R, Sharshar T, et al. Exercise-induced depression of the diaphragm motor evoked potential is not affected by non-invasive ventilation. *Respir Physiol Neurobiol* 2007;155(3):243–54. <https://doi.org/10.1016/j.resp.2006.06.007>.
- Demoule A, Verin E, Locher C, Derenne J-P, Similowski T. Validation of surface recordings of the diaphragm response to transcranial magnetic stimulation in humans. *J Appl Physiol* 2003;94(2):453–61. <https://doi.org/10.1152/japplphysiol.00581.2002>.
- Groppa S, Oliviero A, Eisen A, Quartarone A, Cohen LG, Mall V, et al. A practical guide to diagnostic transcranial magnetic stimulation: Report of an IFCN committee. *Clin Neurophysiol* 2012;123(5):858–82. <https://doi.org/10.1016/j.clinph.2012.01.010>.
- Halawa I, Reichert K, Aberka AS, Sommer M, Peterchev AV, Paulus W. Effect of Pulse Duration and Direction on Plasticity Induced by 5 Hz Repetitive Transcranial Magnetic Stimulation in Correlation With Neuronal Depolarization. *Front Neurosci* 2021;15. <https://doi.org/10.3389/FNINS.2021.773792>.
- Hopkinson NS, Sharshar T, Dayer MJ, Lofaso F, Moxham J, Polkey MI. The effect of acute non-invasive ventilation on corticospinal pathways to the respiratory muscles in chronic obstructive pulmonary disease. *Respir Physiol Neurobiol* 2012;183(1):41–7. <https://doi.org/10.1016/j.resp.2012.05.018>.
- Hopkinson NS, Sharshar T, Ross ET, Nickol AH, Dayer MJ, Porcher R, et al. Corticospinal control of respiratory muscles in chronic obstructive pulmonary disease. *Respir Physiol Neurobiol* 2004;141(1):1–12. <https://doi.org/10.1016/j.resp.2004.04.003>.
- Kammer T, Beck S, Thielscher A, Laubis-Herrmann U, Topka H. Motor thresholds in humans: a transcranial magnetic stimulation study comparing different pulse waveforms, current directions and stimulator types. *Clin Neurophysiol* 2001;112(2):250–8. [https://doi.org/10.1016/S1388-2457\(00\)00513-7](https://doi.org/10.1016/S1388-2457(00)00513-7).
- Kawaguchi Y, Kitagawa H, Nakamura H, Gejo R, Kimura T. Neurophysiological tests of respiratory function by compound muscle action potentials from the diaphragm. *J Bone Jt Surg - Ser B* 2000;82-B(5):695–701.
- Khedr EM, El Shinawy O, Khedr T, Aziz Ali YA, Awad EM. Assessment of corticodiaphragmatic pathway and pulmonary function in acute ischemic stroke patients. *Eur J Neurol* 2000;7(3):323–30. <https://doi.org/10.1046/j.1468-1331.2000.00078.x>.
- Khedr EM, Trakhan MN. Localization of diaphragm motor cortical representation and determination of corticodiaphragmatic latencies by using magnetic stimulation in normal adult human subjects. *Eur J Appl Physiol* 2001;85:560–6. <https://doi.org/10.1007/s004210100504>.
- Laguénay A, Arnaud A, Le Masson G, Burbaud P, Deliaç P, Marthan R. Study of central and peripheral conduction to the diaphragm in 22 patients with definite multiple sclerosis. *Electromyogr Clin Neurophysiol* 1998;38:333–42.
- Lavolette L, Nierat M-C, Hudson AL, Raux M, Allard E, Similowski T, et al. The Supplementary Motor Area Exerts a Tonic Excitatory Influence on Corticospinal Projections to Phrenic Motoneurons in Awake Humans. *PLoS ONE* 2013;8(4):e62258. <https://doi.org/10.1371/journal.pone.0062258>.
- Lazzaro VD, Oliviero A, Saturno E, Pilato F, Insola A, Mazzone P, Profice P, et al. The effect on corticospinal volleys of reversing the direction of current induced in the motor cortex by transcranial magnetic stimulation. *Exp Brain Res* 2001;138(2):268–73.
- Leflaucheur J-P, Lofaso F. Diaphragmatic silent period to transcranial magnetic cortical stimulation for assessing cortical motor control of the diaphragm. *Exp Brain Res* 2002;146(3):404–9. <https://doi.org/10.1007/s00221-002-1197-3>.
- Lissens MA. Motor evoked potentials of the human diaphragm elicited through magnetic transcranial brain stimulation. *J Neurol Sci* 1994;124(2):204–7. [https://doi.org/10.1016/0022-510X\(94\)90327-1](https://doi.org/10.1016/0022-510X(94)90327-1).
- Lissens MA, Vanderstraeten GG. Motor evoked potentials of the respiratory muscles in tetraplegic patients. *Spinal Cord* 1996;34(11):673–8. <https://doi.org/10.1038/sc.1996.122>.
- Maccabee PJ, Nagarajan SS, Amassian VE, Durand DM, Szabo AZ, Ahad AB, et al. Influence of pulse sequence, polarity and amplitude on magnetic stimulation of human and porcine peripheral nerve. *J Physiol* 1998;513(2):571–85.
- Maskill D, Murphy K, Mier A, Owen M, Guz A. Motor cortical representation of the diaphragm in man. *J Physiol* 1991;443:105–21. <https://doi.org/10.1113/jphysiol.1991.sp018825>.
- Miscio G, Gukov B, Pisano F, Mazzini L, Baudo S, Salvadori A, et al. The corticodiaphragmatic pathway involvement in amyotrophic lateral sclerosis: Neurophysiological, respiratory and clinical considerations. *J Neurol Sci* 2006;251(1–2):10–6. <https://doi.org/10.1016/j.jns.2006.05.059>.
- Murphy K, Mier A, Adams L, Guz A. Putative cerebral cortical involvement in the ventilatory response to inhaled CO₂ in conscious man. *J Physiol* 1990;420:1–18. <https://doi.org/10.1113/jphysiol.1990.sp017898>.
- Niehaus L, Meyer BU, Weyh T. Influence of pulse configuration and direction of coil current on excitatory effects of magnetic motor cortex and nerve stimulation. *Clin Neurophysiol* 2000;111:75–80. [https://doi.org/10.1016/S1388-2457\(99\)00198-4](https://doi.org/10.1016/S1388-2457(99)00198-4).
- Peterchev AV, Goetz SM, Westin GG, Luber B, Lisanby SH. Pulse width dependence of motor threshold and input-output curve characterized with controllable pulse parameter transcranial magnetic stimulation. *Clin Neurophysiol* 2013;124:1364–72. <https://doi.org/10.1016/j.clinph.2013.01.011>.
- Rosler KM, Nirikko AC, Rihs F, Hess CW. Motor-evoked responses to transcranial brain stimulation persist during cataplexy: A case report. *Sleep* 1994;17:168–71. <https://doi.org/10.1093/sleep/17.2.168>.
- Rossi S, Antal A, Bestmann S, Bikson M, Brewer C, Brockmüller J, et al. Safety and recommendations for TMS use in healthy subjects and patient populations, with updates on training, ethical and regulatory issues: Expert Guidelines. *Clin Neurophysiol* 2021;132:269–306. <https://doi.org/10.1016/j.clinph.2020.10.003>.
- Rossini PM, Barker AT, Berardelli A, Caramia MD, Caruso G, Cracco RQ, et al. Non-invasive electrical and magnetic stimulation of the brain, spinal cord and roots: basic principles and procedures for routine clinical application. Report of an IFCN committee. *Electroencephalogr Clin Neurophysiol* 1994;91:79–92. [https://doi.org/10.1016/0013-4694\(94\)90029-9](https://doi.org/10.1016/0013-4694(94)90029-9).
- Rossini PM, Burke D, Chen R, Cohen LG, Daskalakis Z, Di Iorio R, et al. Non-invasive electrical and magnetic stimulation of the brain, spinal cord, roots and peripheral nerves: Basic principles and procedures for routine clinical and research application: An updated report from an I.F.C.N. Committee. *Clin Neurophysiol* 2015;126:1071–107. <https://doi.org/10.1016/j.clinph.2015.02.001>.
- Ruohonen J, Karhu J. Navigated transcranial magnetic stimulation. *Neurophysiol Clin* 2010;40:7–17. <https://doi.org/10.1016/j.neucli.2010.01.006>.
- Sakai K, Ugawa Y, Terao Y, Hanajima R, Furubayashi T, Kanazawa I. Preferential activation of different I waves by transcranial magnetic stimulation with a figure-of-eight-shaped coil. *Exp Brain Res* 1997;113:24–32. <https://doi.org/10.1007/BF02454139>.
- Sharshar T, Hopkinson NS, Jonville S, Prigent H, Carlier R, Dayer MJ, et al. Demonstration of a second rapidly conducting cortico-diaphragmatic pathway in humans. *J Physiol* 2004a;560:897–908. <https://doi.org/10.1113/jphysiol.2004.061150>.
- Sharshar T, Ross E, Hopkinson NS, Dayer M, Nickol A, Lofaso F, et al. Effect of voluntary facilitation on the diaphragmatic response to transcranial magnetic stimulation. *J Appl Physiol* 2003;95:26–34. <https://doi.org/10.1152/japplphysiol.00918.2002>.
- Sharshar T, Ross ET, Hopkinson NS, Porcher R, Nickol AH, Jonville S, et al. Depression of diaphragm motor cortex excitability during mechanical ventilation. *J Appl Physiol* 2004b;97:3–10. <https://doi.org/10.1152/japplphysiol.01099.2003>.
- Shirota Y, Sommer M, Paulus W. Strength-Duration Relationship in Paired-pulse Transcranial Magnetic Stimulation (TMS) and Its Implications for Repetitive TMS. *Brain Stimul* 2016;9:755–61. <https://doi.org/10.1016/j.brs.2016.04.019>.
- Similowski T, Attali V, Bensimon G, Salachas F, Mehiri S, Arnulf I, et al. Diaphragmatic dysfunction and dyspnoea in amyotrophic lateral sclerosis. *Eur Respir J* 2000;15:332–7. <https://doi.org/10.1034/j.1399-3003.2000.15b19.x>.
- Similowski T, Catala M, Rancurel G, Derenne JP. Impairment of central motor conduction to the diaphragm in stroke. *Am J Respir Crit Care Med* 1996a;154:436–41. <https://doi.org/10.1164/airccm.154.2.8756819>.
- Similowski T, Duguet A, Straus C, Attali V, Boisteau D, Derenne JP. Assessment of the voluntary activation of the diaphragm using cervical and cortical magnetic stimulation. *Eur Respir J* 1996b;9:1224–31. <https://doi.org/10.1183/09031936.96.09061224>.
- Similowski T, Straus C, Attali V, Duguet A, Jourdain B, Derenne JP. Assessment of the motor pathway to the diaphragm using cortical and cervical magnetic stimulation in the decision-making process of phrenic pacing. *Chest* 1996c;110:1551–7. <https://doi.org/10.1378/chest.110.6.1551>.
- Similowski T, Straus C, Coic L, Derenne JP. Facilitation-independent response of the diaphragm to cortical magnetic stimulation. *Am J Respir Crit Care Med* 1996d;154:1771–7. <https://doi.org/10.1164/airccm.154.6.8970369>.
- Sommer M, Alfaro A, Rummel M, Speck S, Lang N, Tings T, et al. Half sine, monophasic and biphasic transcranial magnetic stimulation of the human motor cortex. *Clin Neurophysiol* 2006;117:838–44. <https://doi.org/10.1016/j.clinph.2005.10.029>.
- Sommer M, Lang N, Tergau F, Paulus W. Neuronal tissue polarization induced by repetitive transcranial magnetic stimulation? *NeuroReport* 2002;13:809–11. <https://doi.org/10.1097/00001756-200205070-00015>.
- Sommer M, Norden C, Schmack L, Rothkegel H, Lang N, Paulus W. Opposite Optimal Current Flow Directions for Induction of Neuroplasticity and Excitation Threshold in the Human Motor Cortex. *Brain Stimul* 2013;6:363–70. <https://doi.org/10.1016/j.brs.2012.07.003>.
- Stephani C, Paulus W, Sommer M. The effect of current flow direction on motor hot spot allocation by transcranial magnetic stimulation. *Physiol Rep* 2016;4. <https://doi.org/10.14814/phy2.12666>.
- Straus C, Locher C, Zelter M, Derenne JP, Similowski T. Facilitation of the diaphragm response to transcranial magnetic stimulation by increases in human respiratory drive. *J Appl Physiol* 2004;97:902–12. <https://doi.org/10.1152/japplphysiol.00989.2003>.
- Terao Y, Ugawa Y, Hanajima R, Machii K, Furubayashi T, Mochizuki H, et al. Predominant activation of I1-waves from the leg motor area by transcranial magnetic stimulation. *Brain Res* 2000;859:137–46. [https://doi.org/10.1016/S0006-8993\(00\)01975-2](https://doi.org/10.1016/S0006-8993(00)01975-2).
- Tings T, Lang N, Tergau F, Paulus W, Sommer M. Orientation-specific fast rTMS maximizes corticospinal inhibition and facilitation. *Exp Brain Res* 2005;164:323–33. <https://doi.org/10.1007/S00221-005-2253-6>.
- Verin E, Ross E, Demoule A, Hopkinson N, Nickol A, Fauroux B, et al. Effects of exhaustive incremental treadmill exercise on diaphragm and quadriceps motor potentials evoked by transcranial magnetic stimulation. *J Appl Physiol* 2004;96:253–9. <https://doi.org/10.1152/japplphysiol.00325.2003>.
- Verin E, Straus C, Demoule A, Mialon P, Derenne JP, Similowski T. Validation of improved recording site to measure phrenic conduction from surface electrodes in humans. *J Appl Physiol* 2002;92:967–74. <https://doi.org/10.1152/japplphysiol.00652.2001>.
- Welch JF, Argento PJ, Mitchell GS, Fox EJ. Reliability of diaphragmatic motor-evoked potentials induced by transcranial magnetic stimulation. *J Appl Physiol* 2020;129:1393–404. <https://doi.org/10.1152/JAPPLPHYSIOL.00486.2020>.

- Welch JF, Perim RR, Argento PJ, Sutor TW, Vose AK, Nair J, et al. Effect of acute intermittent hypoxia on cortico-diaphragmatic conduction in healthy humans. *Exp Neurol* 2021;339. <https://doi.org/10.1016/j.expneurol.2021.113651>.
- Zifko U, Remtulla H, Power K, Harker L, Bolton CF. Transcortical and cervical magnetic stimulation with recording of the diaphragm. *Muscle Nerve* 1996;19:614–20. 10.1002/(SICI)1097-4598(199605)19:5<614::AID-MUS9>3.0.CO;2-E.
- Zmeykina E, Mittner M, Paulus W, Turi Z. Weak rTMS-induced electric fields produce neural entrainment in humans. *Sci Rep* 2020;10. <https://doi.org/10.1038/s41598-020-68687-8>.